Disease and immunity

WHEN THE BODY IS ATTACKED BY DISEASE-CAUSING ORGANISMS, IT HAS A RANGE OF RESPONSES.

SEE ALSO
❮ 24–25 Cells at work
❮ 26–27 Fungi and single-celled life
Body systems 62–63 ❯

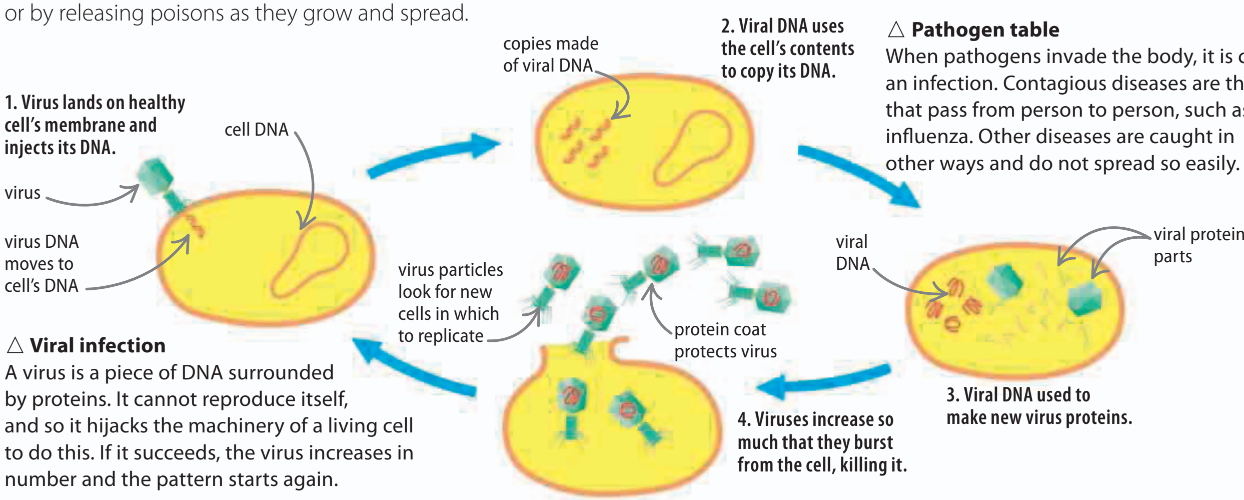
The immune system is a highly complex defense system that looks for, and then destroys, foreign bodies that get inside the body. These foreign bodies use the body as a place to live and reproduce, which can cause illness.

Pathogens

The agents that cause disease are called pathogens. Most are living organisms, such as bacteria (often called germs), but illnesses are also caused by viruses, which are not generally considered to be living. The pathogens infect body tissues, and cause symptoms by killing cells, or by releasing poisons as they grow and spread.

Infection name	Type of pathogen	What it does	Symptom
streptococcus	bacterium	lives on skin and throat	sore throat
plasmodium	protist	kills body cells	malaria
threadworm	nematode worm	lives in intestines	itchy anus
H1N1	virus	invades body cells	fever and joint pain

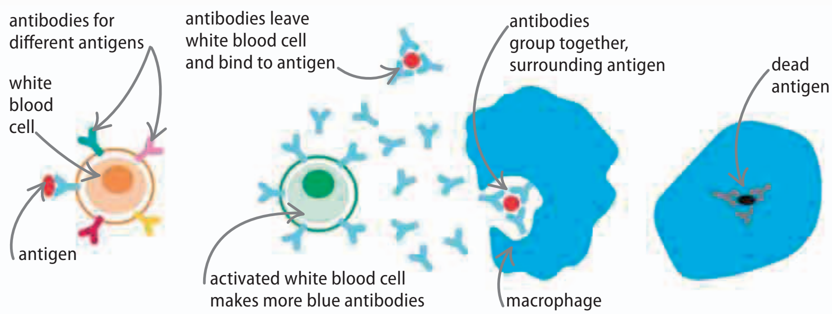
△ **Pathogen table**
When pathogens invade the body, it is called an infection. Contagious diseases are those that pass from person to person, such as influenza. Other diseases are caught in other ways and do not spread so easily.



△ **Viral infection**
A virus is a piece of DNA surrounded by proteins. It cannot reproduce itself, and so it hijacks the machinery of a living cell to do this. If it succeeds, the virus increases in number and the pattern starts again.

White blood cells

White blood cells are the detectives of the body, patrolling the bloodstream looking for invaders. When they find one, the white blood cells copy the invading pathogen's antigen—the chemical marker on its surface. Then, the blood cells generate a protein, called an antibody, that flags the attacker for removal from the body. Amazingly, the immune system remembers the antibodies for all past attacks, and so can only be infected once by the same pathogen.



△ **1. Seeking**
The white blood cell recognizes the antigen on an object in the body as being foreign.

△ **2. Attacking**
Antibodies are released, and large white blood cells called macrophages engulf the antigen.

△ **3. Destroying**
Destructive enzymes called lysosomes finally kill the antigen inside the macrophage.